

Department of Artificial Intelligence & Data Science
School of Computing, FAST-NUCES

Degree: DS/AI

Total Marks: 20

Instructor: Dr. Zohair Ahmed

Due Date: As Per GCR

Assignment 3: Experimentation and Expansion

Objective

Encourage **original research thinking** by extending the reproduced work with new experiments. Students should **build on insights from Assignment 1 and 2**.

Tasks

1. Research Proposal

Propose **one or more improvements**, such as:

- Model architecture modification
 - New loss function
 - Additional training data
 - Cross-domain evaluation
 - Efficiency improvements
 - Multimodal extension
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2. Experimental Design

Students must clearly define:

- Hypothesis
 - Experimental setup
 - Baselines
 - Evaluation metrics
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3. Conduct Experiments

Run experiments and compare:

- Original model
- Reproduced baseline
- Proposed method (Optional Bonus) or Atleast one More Additional Dataset (Must)

4. Results & Analysis

Include:

- Performance comparison tables
- Visualizations (graphs, confusion matrices)
- Statistical significance tests (if applicable)

5. Research Discussion

Explain:

- What improvements were achieved
- Why the modification works (or fails)
- Limitations of the proposed approach

Deliverables

Report Length: 10–12 pages

Structure:

1. Introduction
2. Background and Paper Summary
3. Reproduction Summary
4. Proposed Method
5. Experimental Setup
6. Results and Analysis
7. Discussion
8. Conclusion
9. References

Students must also submit:

- **Updated code repository**
 - **Experiment documentation**
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Evaluation Criteria (Total 100)

Criteria	Weight
Originality of idea	30%
Experimental design	25%
Quality of results	20%
Research discussion	15%
Code & reproducibility	10%